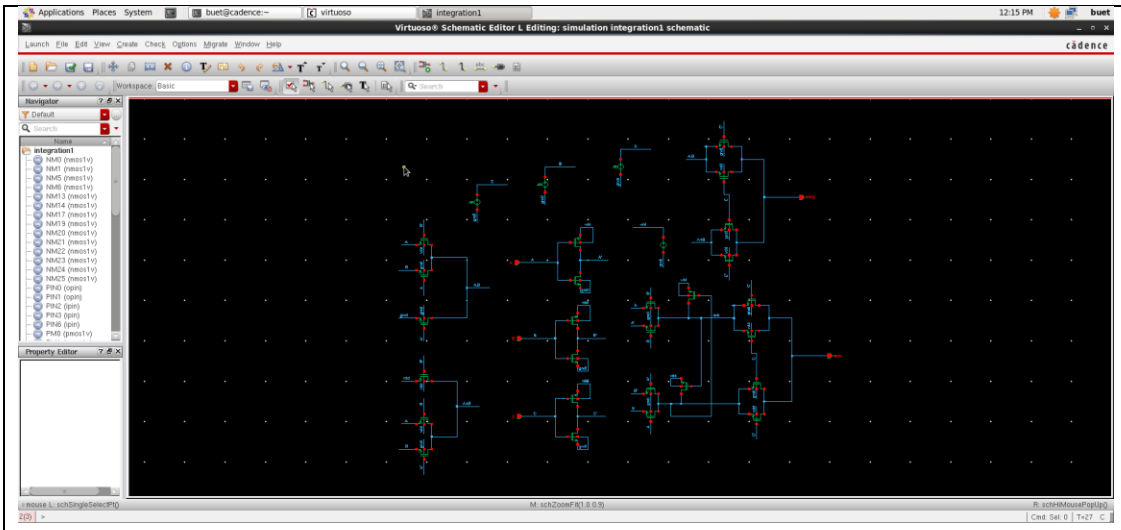
Name of the paper: CMOS Full-Adders for Energy-Efficient Arithmetic Applications

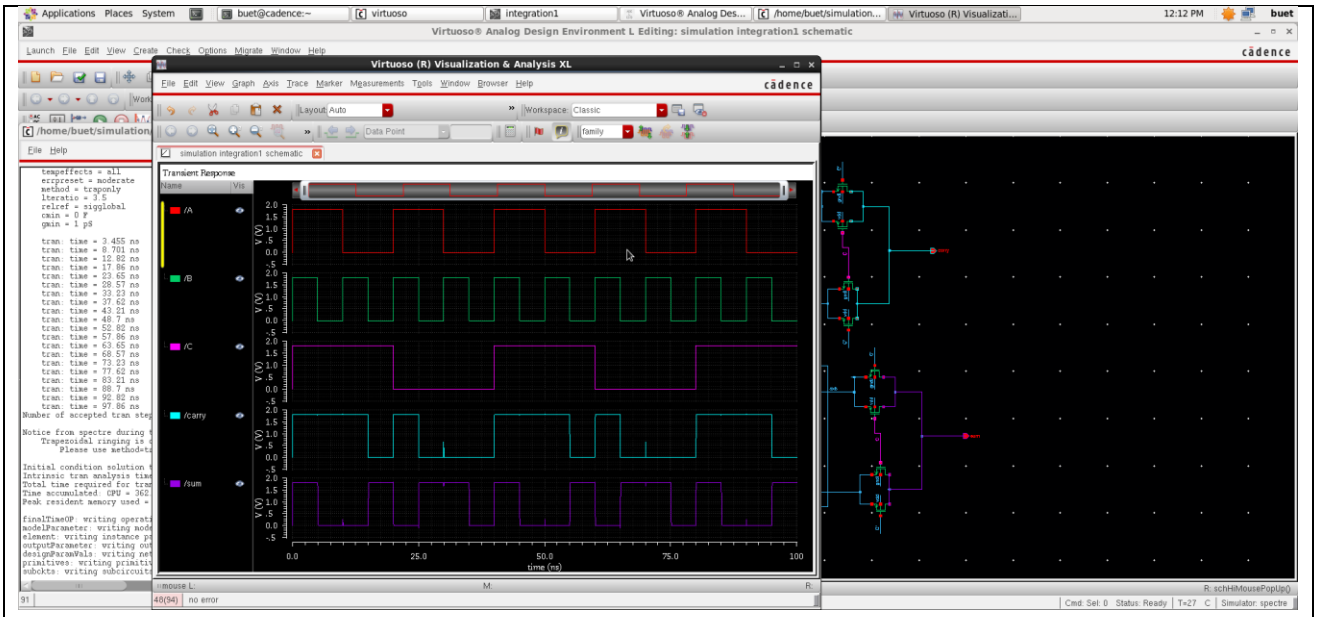
Journal: TVLSI(IEEE-2011)

Symmetrical:



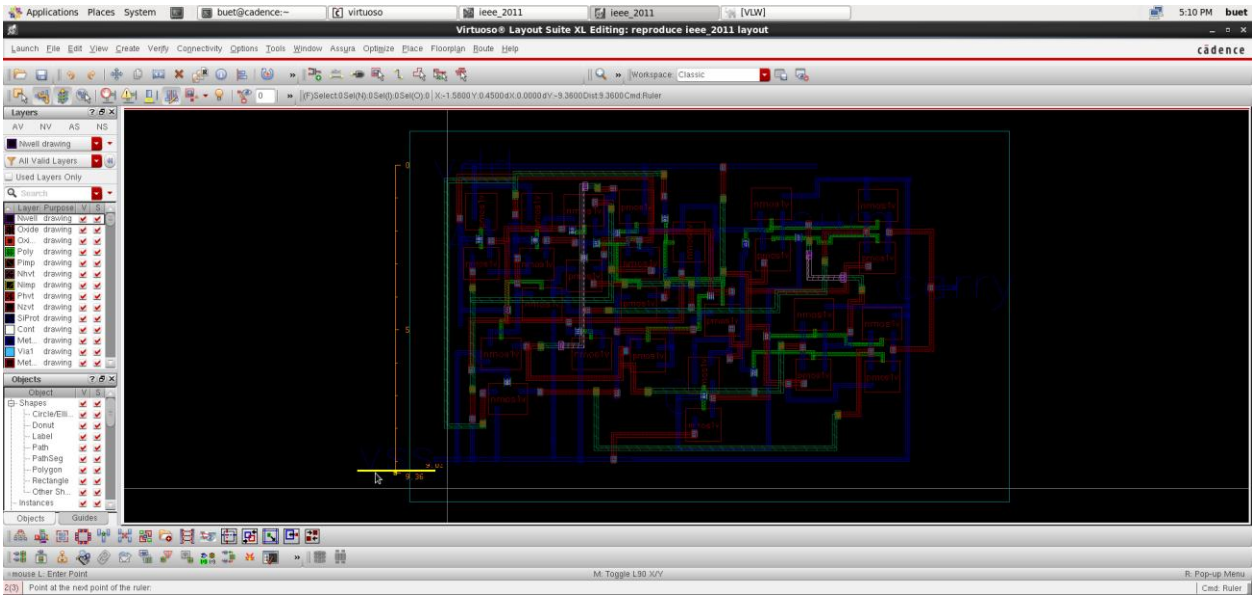
In paper,They produced this result using 180nm technology, but I produced it in 90nm due to its availability.

Output:



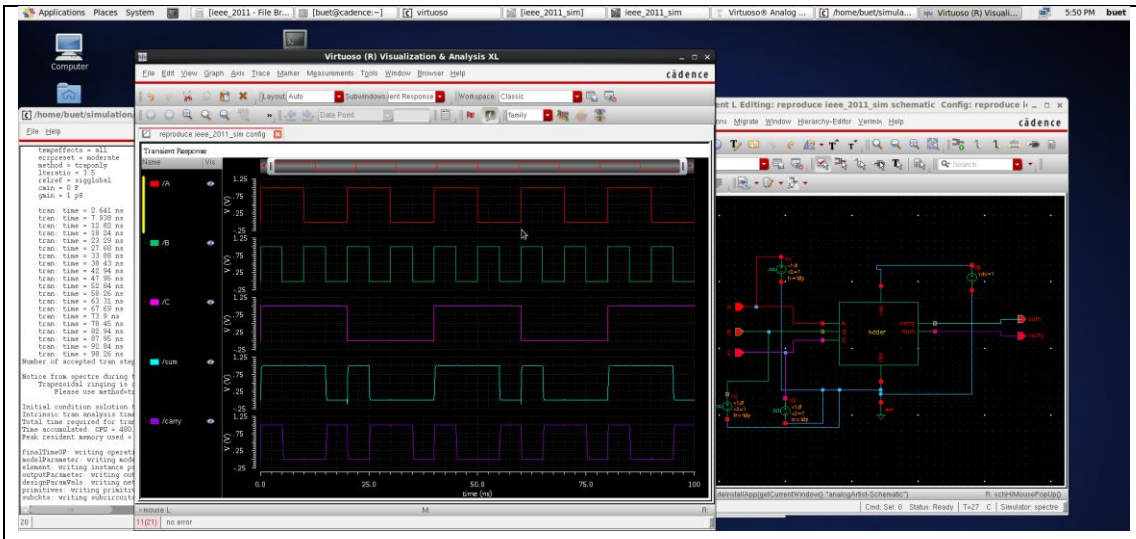
Full voltage swing.
Applied voltage:1v POWER-705.3E-9 Delay-59.93E-12
w/l ratio according to the paper

Layout:



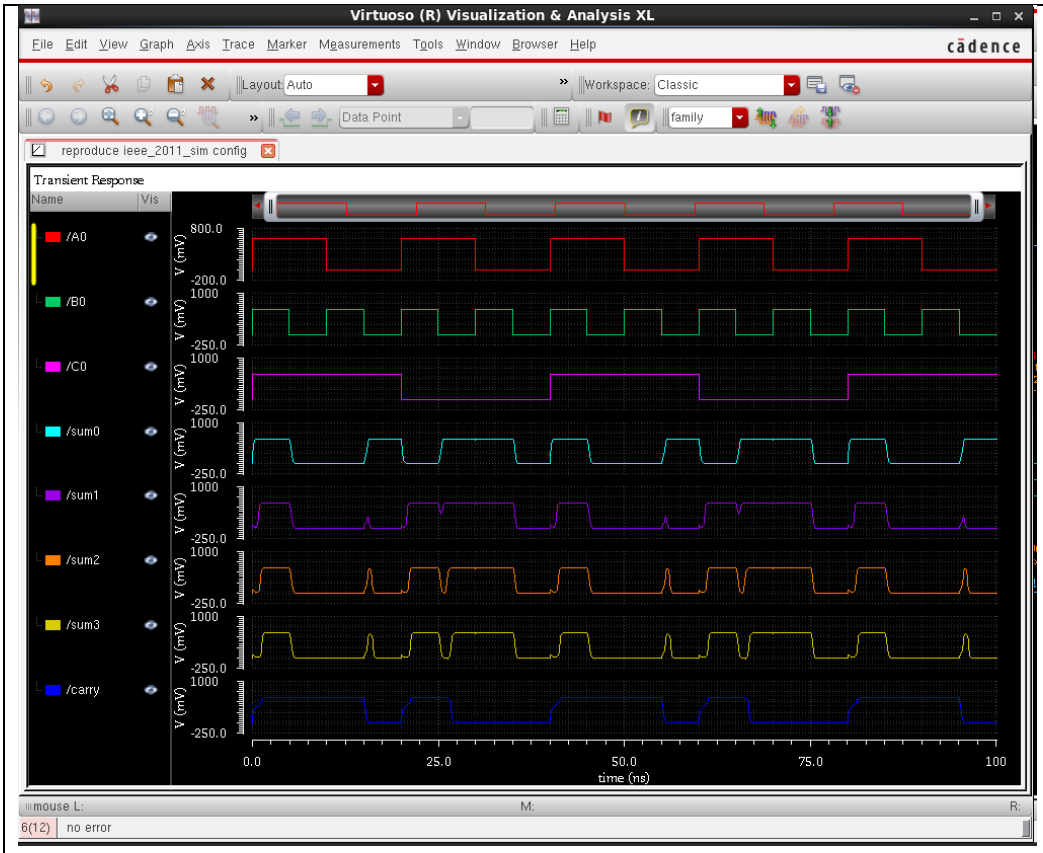
Reproduce Area=9.03*15.47 (90 nm Technology)
In paper =15.1*16*1 (180 nm Technology)

Post layout simulation-



When applied voltage 1V
Power-1.454E-6
133.0E-12

4bit-

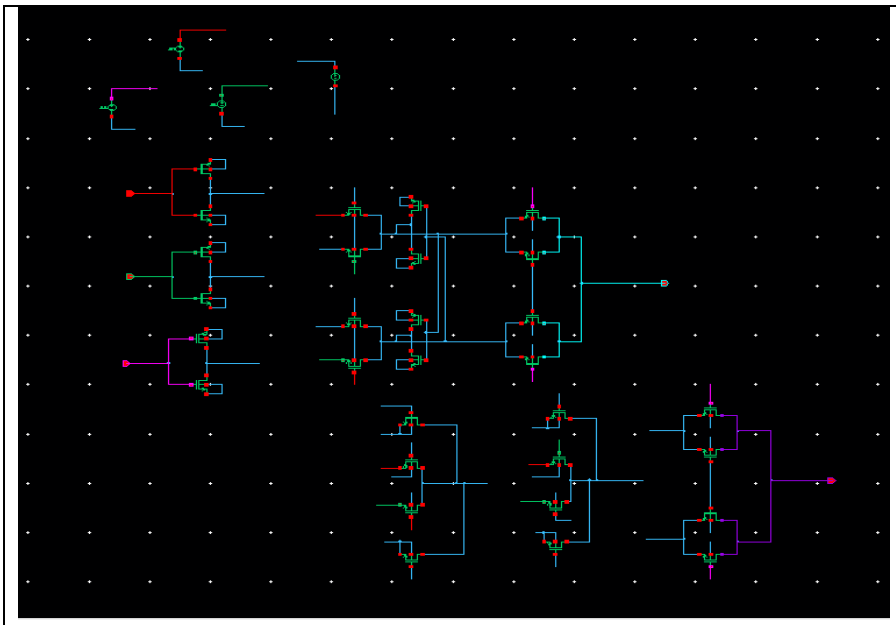


Output wave form of 4 bit operation without load
When applied voltage 0.6v
power-4.019E-9
Delay-265.0E-12
When applied voltage 1v
Power-7.125E-6
Delay-108.0E-12

Name of the paper: CMOS full adder cells based on modified full swing restored complementary pass transistor logic for energy efficient high speed arithmetic applications

Journal: Elsevier (integration-2024)

Symmetrical:

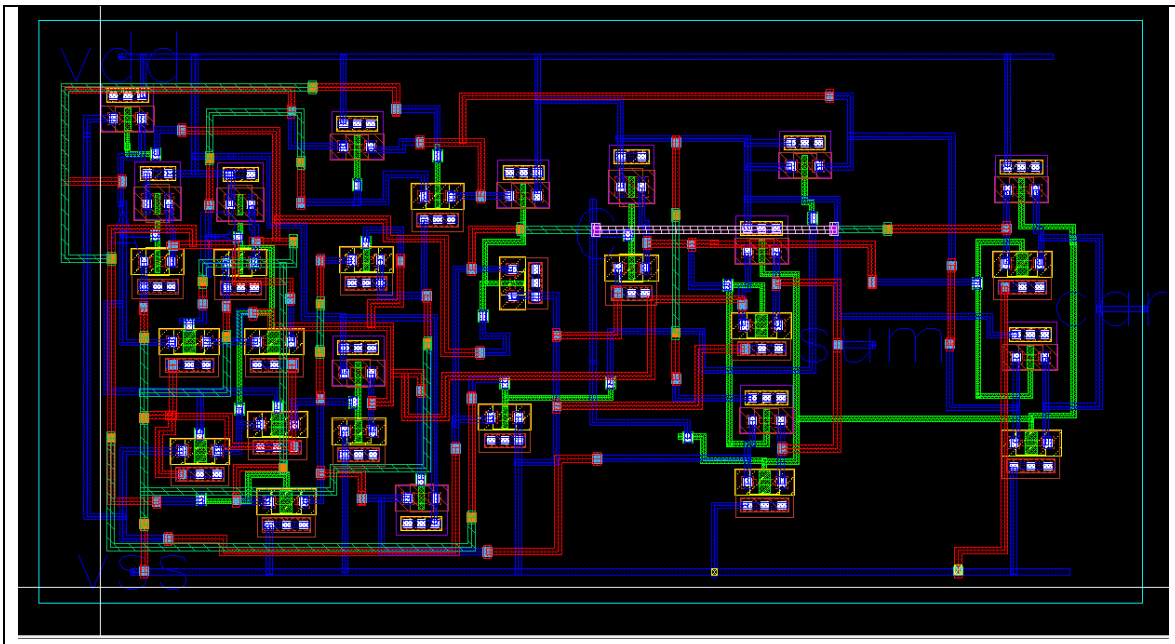


w/l ratio according to paper

Output:

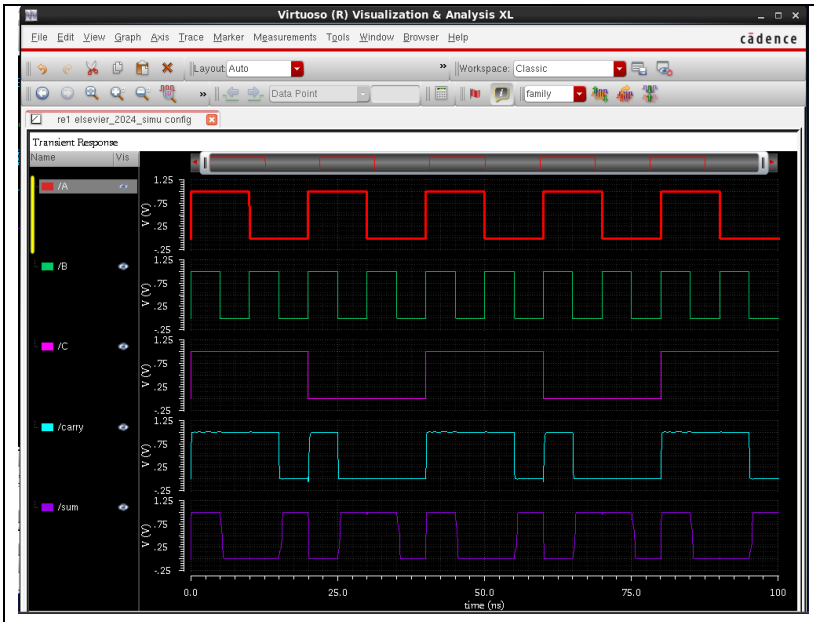
Waveform
Applied voltage -1v

Layout:



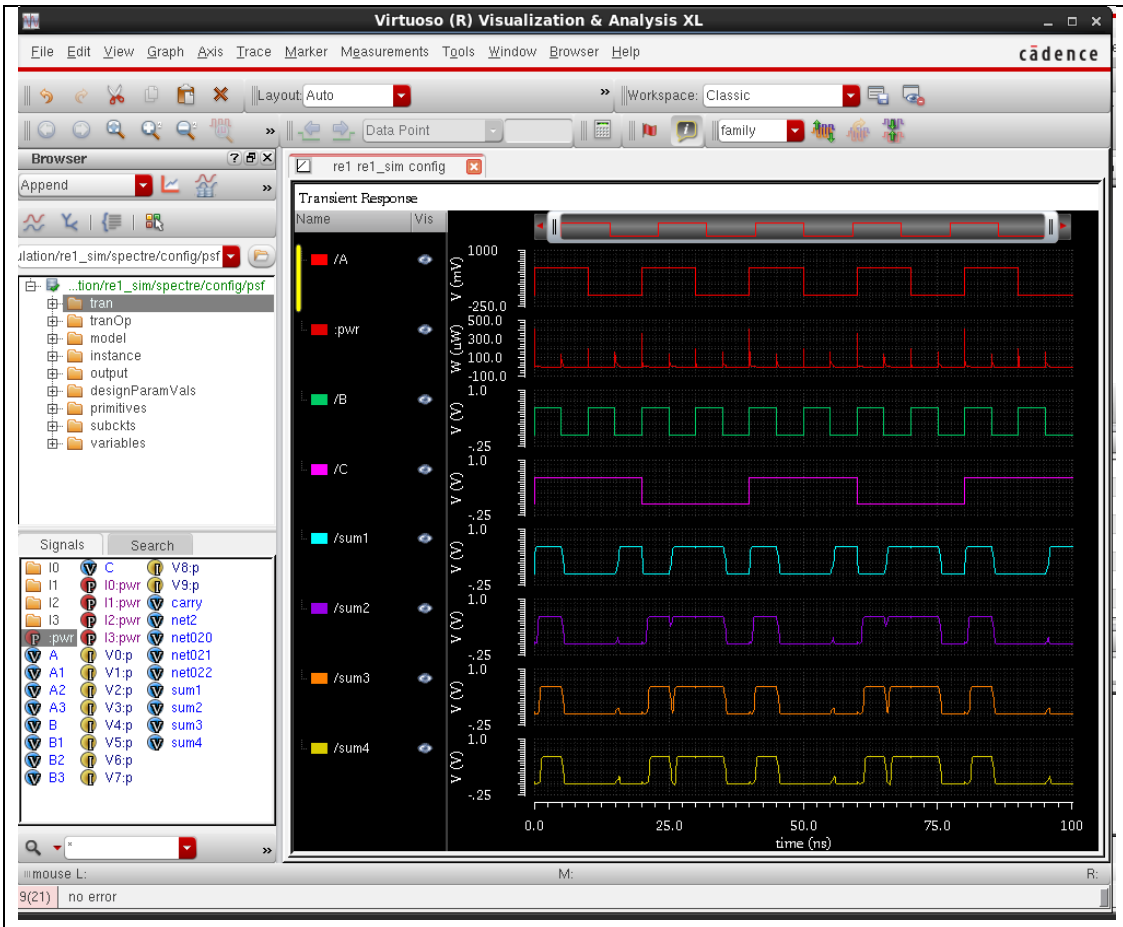
Area=10.79*22.31
In paper , area=23.54*36.48

Post layout result-



When applied voltage-
Power- 3.167E-6
Delay- 85.82E-12

4 bit operation:

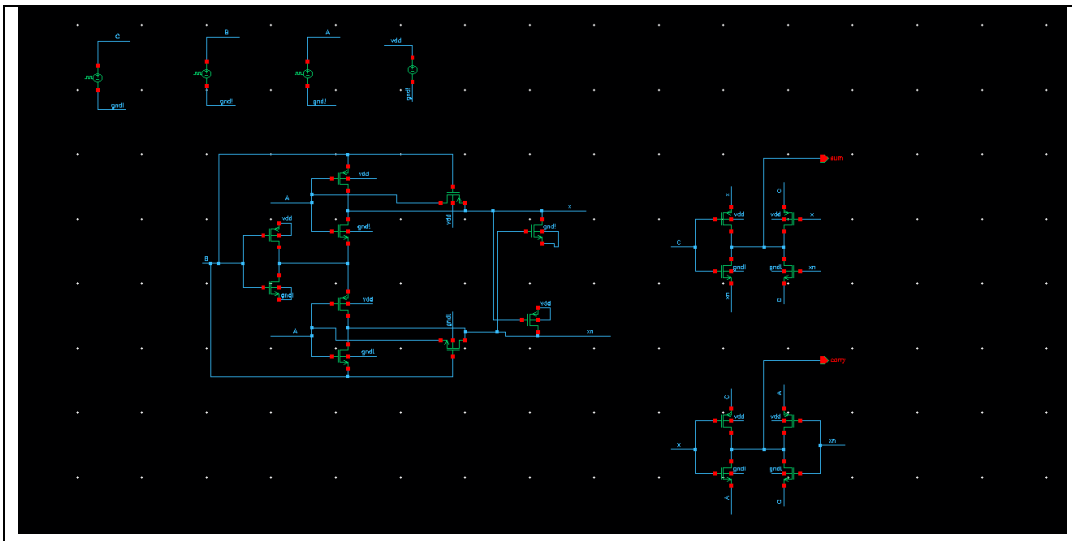


When applied voltage-0.6v
Power-5.75E-6
Delay- 346.9E-12
When applied voltage-1v
Power- 13.81E-6
Delay- 148.1E-12

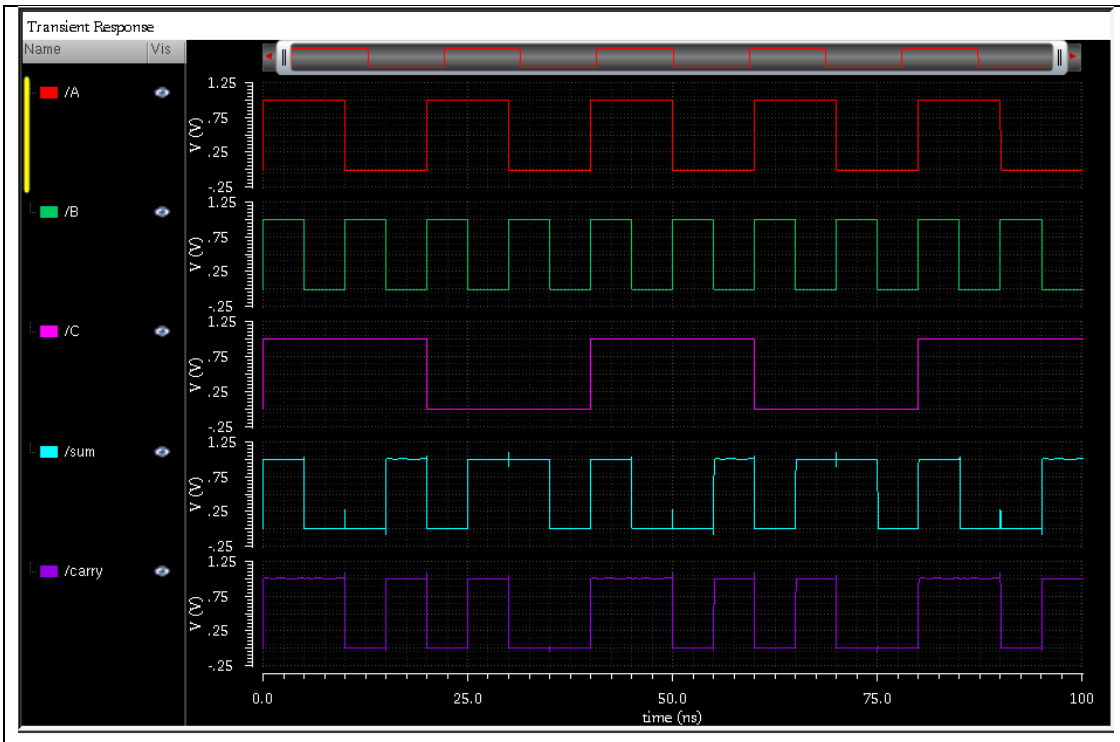
Name of the paper: Design and analysis of a novel low-power and energy-efficient 18T hybrid full adder

Journal: Elsevier (Microelectronics Journal-2018)

Symmetrical:

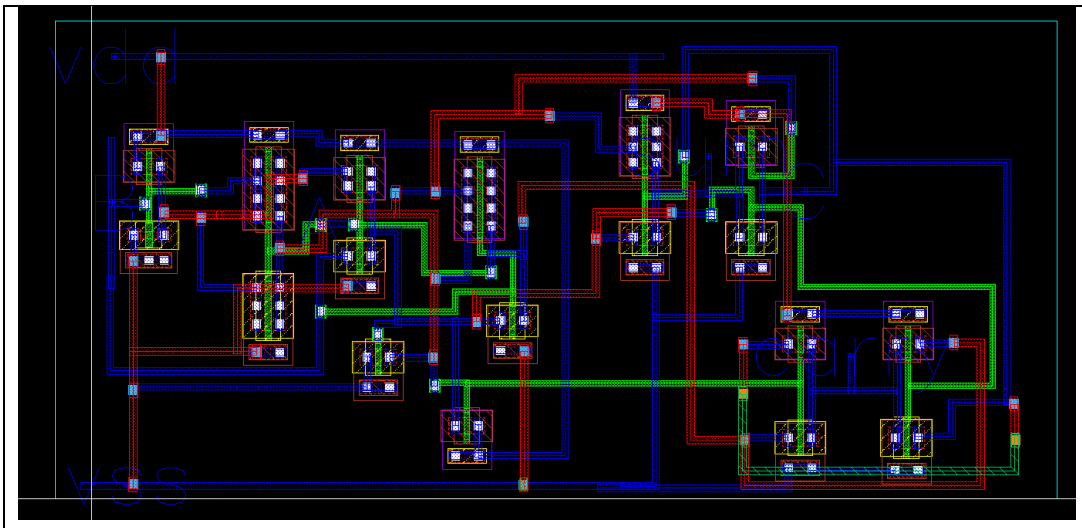


Simulation:



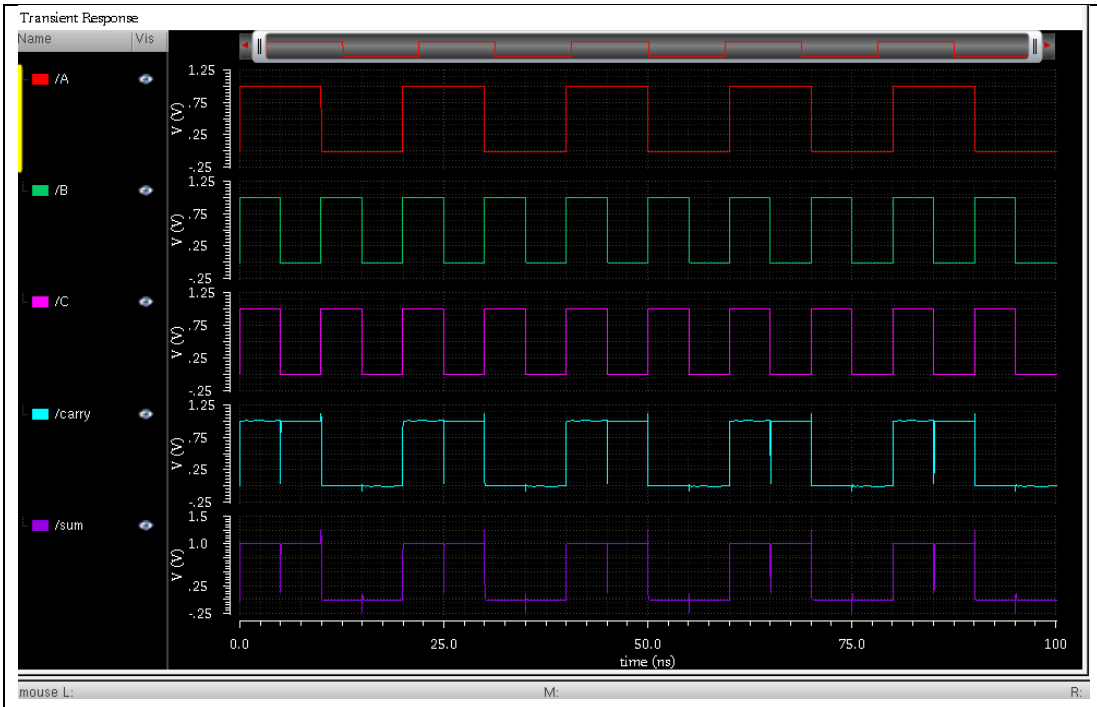
Waveform applied voltage-1
w/l ratio according to paper. Power- 1.356E-6 delay- 36.54E-12

Layout:



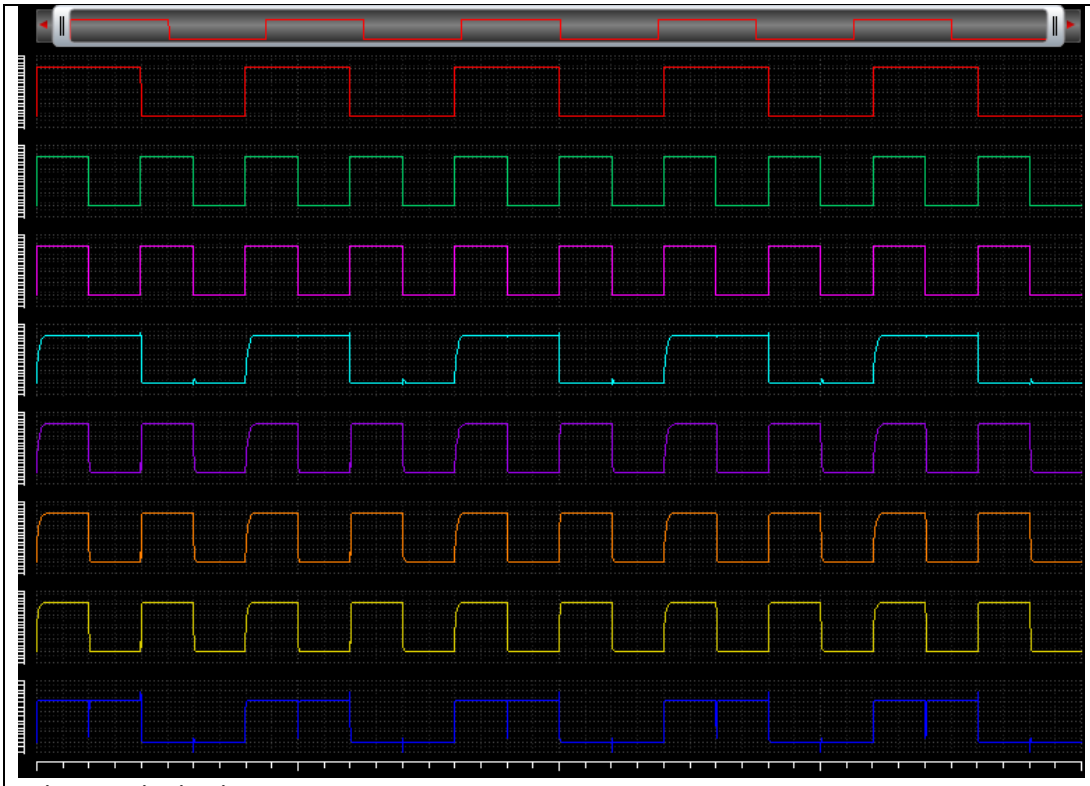
Area=17.17*7.93

Post layout simulation:



Power= 692.4E-9
Delay= 5.014E-9

4bit-simulation:



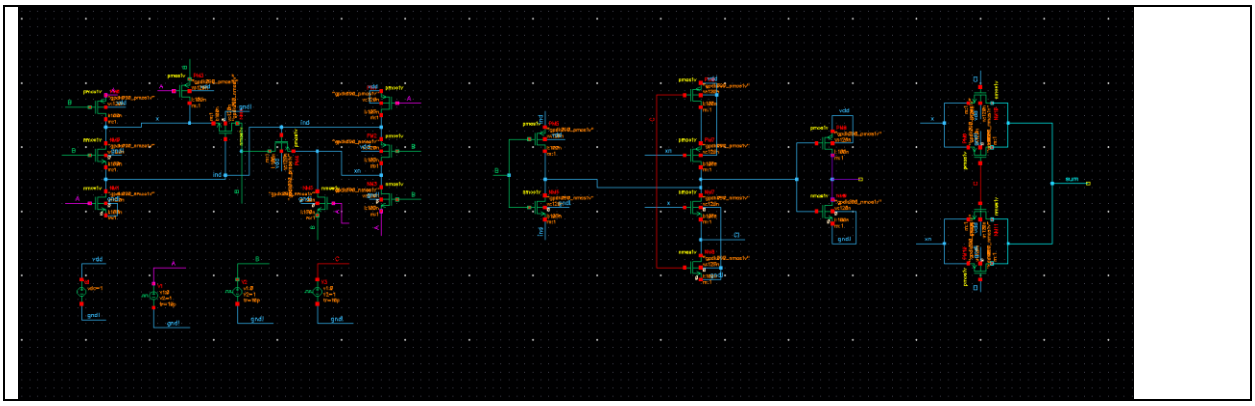
When applied voltage-1v
Power- 2.598E-6
delay- 66.07E-12
PDP- 171.65 aJ

when applied voltage -0.6
power- 845.7E-9
delay- 541.4E-12
PDP- 457.86 aJ

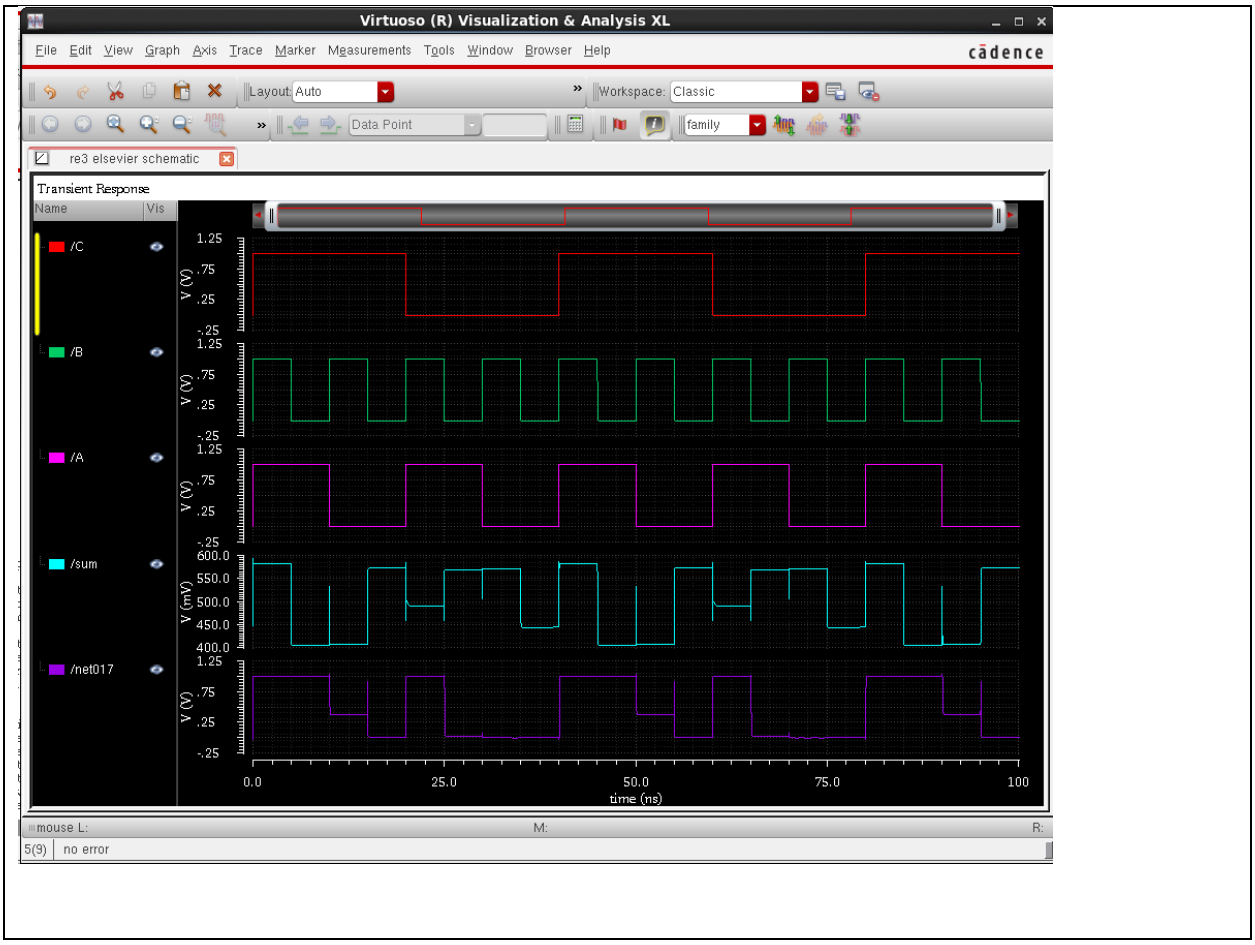
Name of the paper: Design and implementation of 20-T hybrid full adder for high-performance arithmetic applications

Journal: Elsevier (Computers and Electrical Engineering-2020)

Symmetrical:



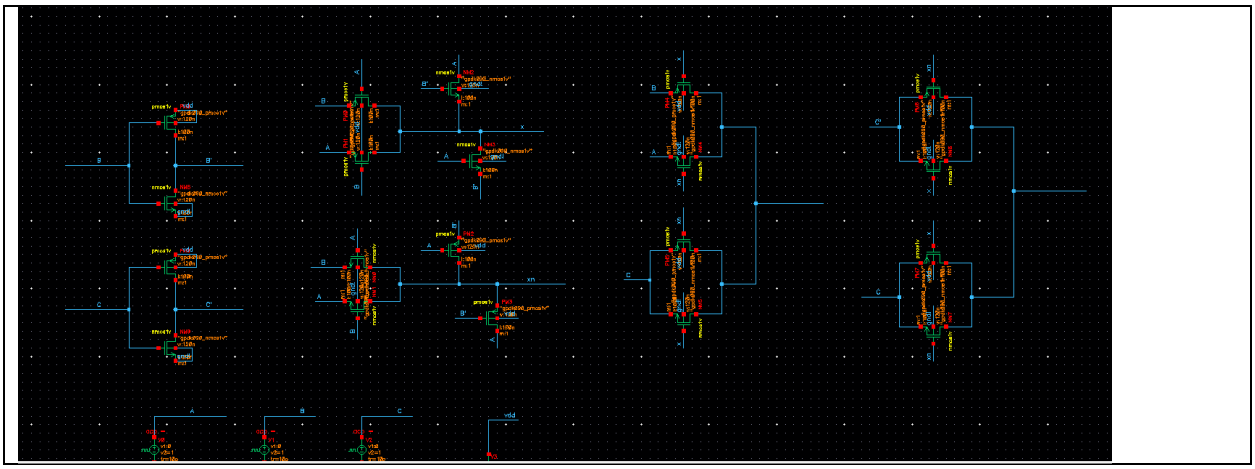
Waveform:



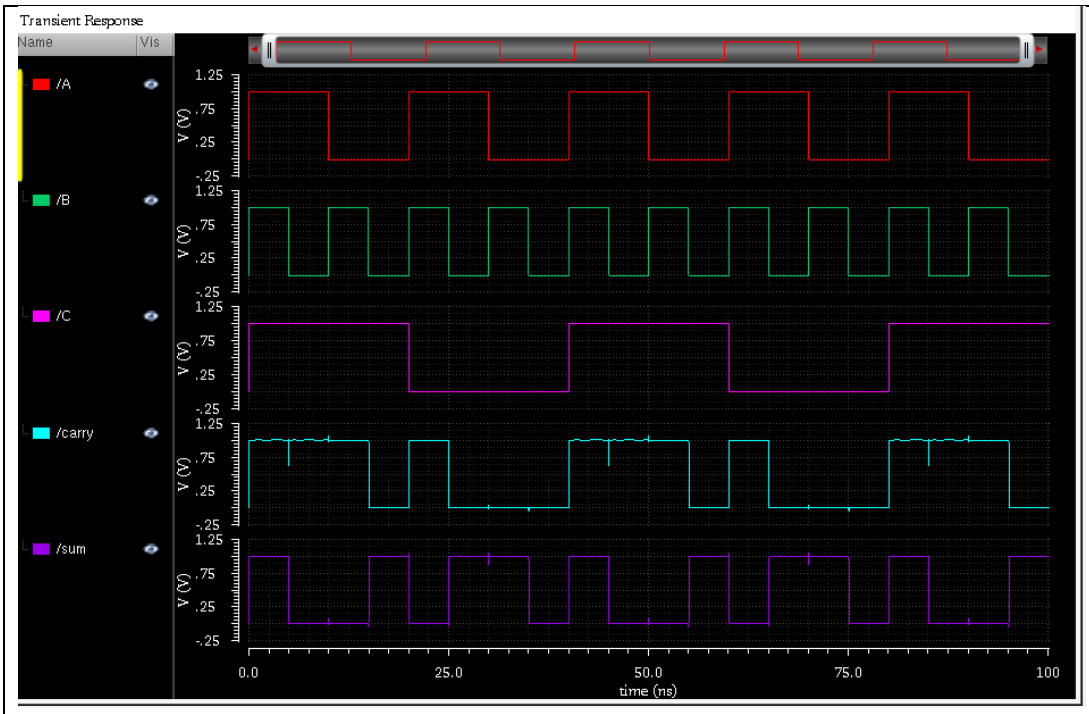
Name of the paper: Design and implementation of 20-T hybrid full adder for high-performance arithmetic applications

Journal: Elsevier (Microelectronics Journal-2021)

Symmetrical:

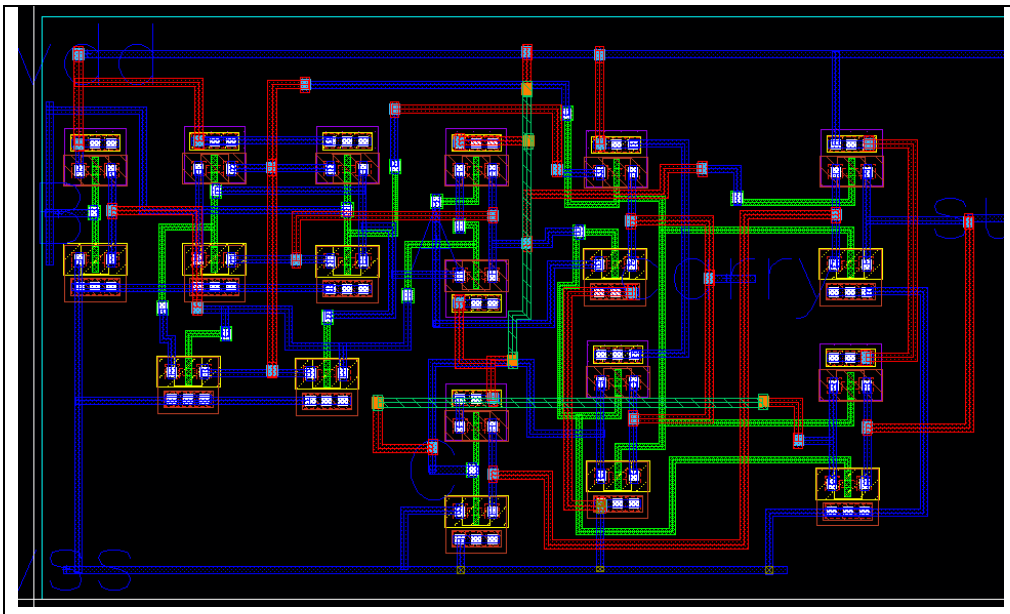


Waveform:



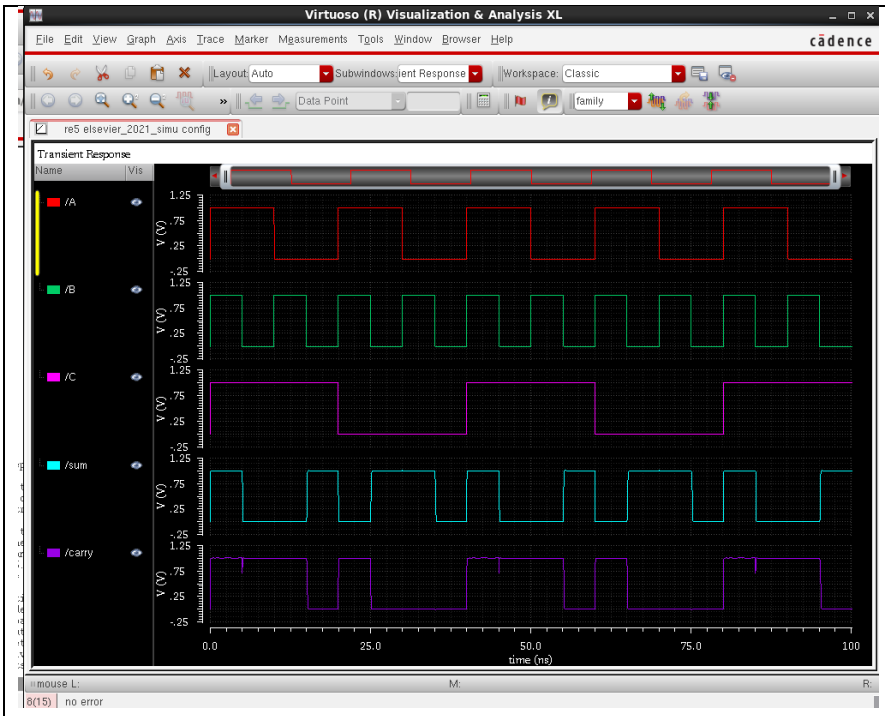
Pre-layout simulation -
Power- 266.4E-9
Delay- 2.575E-12

Layout-



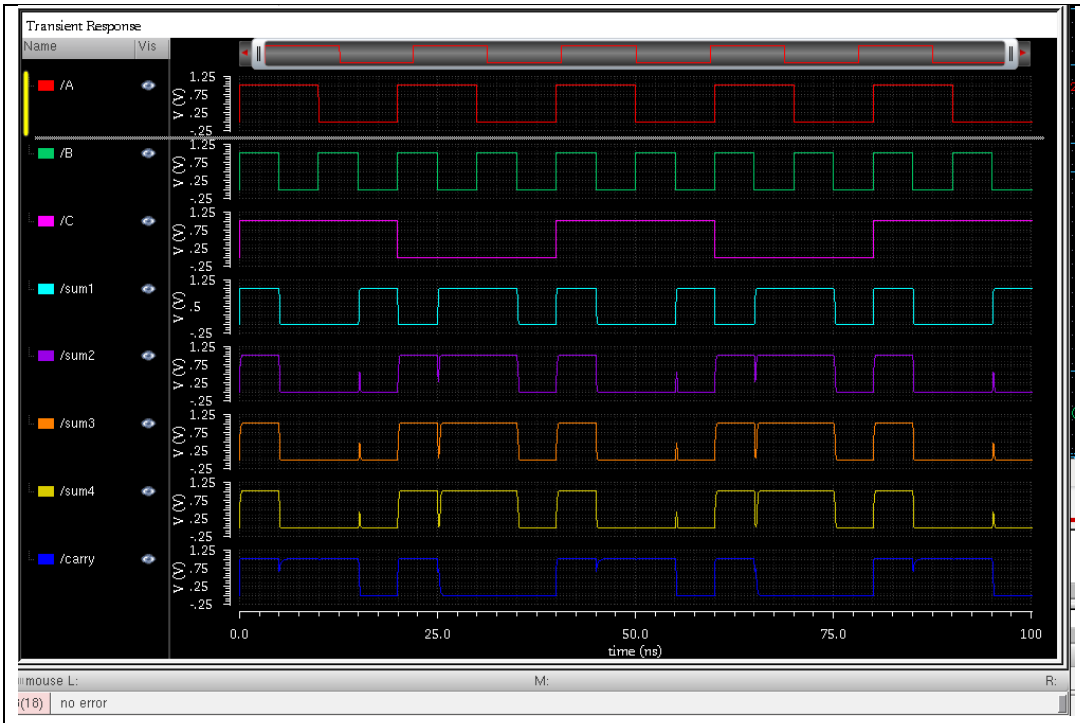
Area=15.8*8.3

Post layout simulation-



Applied voltage-1V
Power- 643.3E-9
Delay- 10.42E-12

4bit-



When applied voltage-1V
Power-2.844E-6
Delay- 11.32E-12

When applied voltage-0.6v
Power-1.548E-6
Delay-32.06E-12